

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

Generated on: 2026-01-14 01:12:21

Total Marks: 100

Examination Paper

Question Q1 (20 marks)

- a) Identify and list all entities in the library scenario. (5)
- b) List all attributes for the identified entities, ensuring to specify any unique constraints. (5)
- c) Draw the ER diagram for the library scenario, including any relationships and cardinalities between the entities: sections, books, and members. [DIAGRAM PLACEHOLDER: Draw the ER diagram for the scenario in the answer booklet.] (5)
- d) Describe the relationships between the library entities and map them to a relational schema including primary and foreign keys. (5)

Question Q2 (15 marks)

- a) Identify the key concepts and definitions related to frame bytes time. (2)
- b) Explain the main components and their relationships related to frame bytes time. (1)
- c) Describe the advantages and disadvantages related to frame bytes time. (2)
- d) Analyze the implementation approach related to frame bytes time. (3)
- e) Design the real-world applications related to frame bytes time. (2)
- f) Calculate the key concepts and definitions related to frame bytes time. (2)
- g) List the main components and their relationships related to frame bytes time. (3)

Question Q3 (25 marks)

- a) Draw an ER diagram representing the library system, illustrating sections, books, copies, and members. Be sure to include all necessary relationships and constraints. (4)
- b) List at least five attributes for the 'Book' entity in your ER diagram, ensuring you include identifiers. (2)
- c) Define the primary key for the 'Section' entity and explain its significance in this database design. (3)

- d) Provide a basic SQL schema for the 'Book' and 'Section' entities including relationships. Ensure all primary keys and foreign keys are specified. (5)
- e) Discuss two potential concurrency issues that may arise when members concurrently attempt to borrow the same book copy and propose solutions for each. (5)
- f) Construct an ER/EER diagram for the scenario described above, showing all entities, relationships, and attributes. (3)
- g) Identify and explain the importance of maintaining a history table for borrowed books and how it will facilitate data auditing. (3)

Question Q4 (40 marks)

- a) Identify the main entities and their attributes. (9)
- b) Explain the relationships between entities and their cardinalities. (5)
- c) Describe how to map the ER diagram to a relational schema. (3)
- d) Analyze the primary and foreign keys in the relational model. (5)
- e) Design the constraints and integrity rules. (5)
- f) Calculate the main entities and their attributes. (8)
- g) List the relationships between entities and their cardinalities. (5)

Diagram Placeholder (ER)

[DIAGRAM PLACEHOLDER: Draw the ER diagram for the scenario in the answer booklet.]